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$$017 \div 6 + 1 = 6$$

So, Q6 and Q1 is solved

Q6: a) If tests can be of any size, decision lists can represent any Boolean function.

Proof:

Any Boolean function can be written as Disjunctive Normal Form (DNF) - as a disjunction of conjunctions of literals.

example: $(R \wedge \neg S \wedge T) \vee (\neg R \wedge \neg S \wedge T) \vee (R \wedge \neg S \wedge \neg T) \vee (R \wedge S \wedge T)$

We can construct a decision list where each test corresponds to one conjunctive clause from the DNF and outputs true.

If no clause matches, we output false (default for DL)

Decision List:

$R \wedge \neg S \wedge T$	$\rightarrow$ True
$\downarrow$ False	
$\neg R \wedge \neg S \wedge T$	$\rightarrow$ True
$\downarrow$ False	
$R \wedge \neg S \wedge \neg T$	$\rightarrow$ True
$\downarrow$ False	
$R \wedge S \wedge T$	$\rightarrow$ True
$\downarrow$ False	
Default	$\rightarrow$ False

Since each DNF clause can include any number of literals, we need tests of arbitrary size to represent them.

If tests can be of any size, we can match any clause in the function.

Therefore, decision lists can represent any Boolean function with arbitrary size.

b) $K\text{-DT} \subseteq K\text{-DL}(n)$

proof:

Each path from the root to leaf in a decision tree of depth $\leq k$ corresponds to a conjunction of at most k literals.

The output at the leaf becomes the decision for that conjunction.

We can construct a k -decision list ($K\text{-DL}$) by converting each such path into a test.

→ the test is the conjunction of literals on the path

→ the decision (output) is the leaf value

→ since each path has $\leq k$ decisions, each test has $\leq k$ literals

Arranging these tests in any order the decision list matches

the same behaviour as the tree.

Therefore every function in k -DT can be represented by a k -DL(n).

Hence, k -DT $\subseteq k$ -DL(n)

c) The number of examples needed for PAC-learning a k -DL(n) function is polynomial in n .

Proof:

In PAC learning, the sample complexity is

$$m \geq \frac{1}{\epsilon} \left(\ln |H| + \ln \frac{1}{\delta} \right)$$

where m is ~~minimum~~ number of examples, H is hypothesis space, ϵ is the error, δ is confidence.

To show m is polynomial in n , we bound $|H|$, the size of the k -DL(n) hypothesis space.

Each test is a conjunction of at most k literals over $2n$ possible literals (with or without negation).

$$\begin{aligned}\text{Number of possible conjunction} &= \sum_{i=0}^k \binom{2n}{i} \\ &= \binom{2n}{0} + \binom{2n}{1} + \dots + \binom{2n}{k} \\ &\leq (2n)^k\end{aligned}$$

Each test has an output true or false.

$$\therefore |H| \leq 2^{(2n)^k} \Rightarrow \ln |H| \leq (2n)^k \ln 2$$

$$\Rightarrow \ln |H| = O(n^k)$$

Thus, $m \geq \frac{1}{\epsilon} \left\{ O(n^k) + \log\left(\frac{1}{\delta}\right) \right\}$; this is polynomial in n .

So, number of examples needed for PAC learning a

k -DL(n) function is polynomial in n .

Q1:

A Boolean function with n inputs ^{maps} each of the 2^n possible input combinations to a single output bit (0 or 1).

Each input combination outputs 0 or 1.

$\therefore$ Total number of Boolean functions = 2^{2^n}

~~$\therefore$ Hypo~~

Lower bound:

Any representation of decision trees must cover at least 2^{2^n} unique functions.

So lower bound of hypothesis space of possible decision trees over n Boolean variables is 2^{2^n} .

Upper bound:

Each decision tree is a binary tree with internal nodes labeled by variables and leaves labeled with outputs (0 or 1).

For n Boolean variables, there are 2^n possible input combinations

In worst case a decision tree might have one leaf for each

input. So maximum number of leaves $= 2^n$.

Let $m = 2^n = \text{max. number of leaves}$.

We know that, a full binary tree with m leaves has exactly $m-1$ internal nodes. So the total number of binary tree shapes is the number of ways to pair up nodes to form a tree.

There are at most $m-1$ internal nodes, each can be labeled with one of the n variables.

$$\therefore \text{Number of ways to assign variables to internal nodes} \\ = n^{m-1} = n^{2^n-1} \leq n^{2^n}$$

There are at most $m=2^n$ leaves, each can be labeled with output 0 or 1.

$$\therefore \text{Number of ways to assign outputs} = 2^m = 2^{2^n}$$

$$\therefore \text{Total number of decision trees} \leq n^{2^n} \times 2^{2^n} = (2n)^{2^n}$$

$\therefore$ Upper bound of the hypothesis space of possible decision trees over n Boolean variables is $(2n)^{2^n}$.